EXPERIMENT - 14

AIM:

To write a code using Raw Sockets. To implement Packet sniffing

CODE:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def Packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        Protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst

        Protocol_name = " "
        if Protocol == 1:
            Protocol_name = "ICMP"
        elif Protocol == 6:
            Protocol_name = "TCP"
        elif protocol == 17:
            Protocol_name = "UDP"
        else:
            Protocol_name = "Unknown Protocol"

    Print(f"Protocol: {Protocol_name}")
    Print(f"Source IP: {src_ip}")
    Print(f"Destination IP: {dst_ip}")
    Print("." * 50)
```

sniff(iface = "wifi", prn = packet
filter = "IP", store = 0)

OUTPUT:

Protocol : TCP
source IP: 57.155.141.117

Destination IP: 192.168.1.6

Protocol : UDP
source IP : 57.155.141.117
Destination IP : 192.168.1.1

Protocol : UDP
Source IP : 57.155.141.117
Destination IP : 192.168.1.1

Protocol : UDP
SourceIP : 57.155.141.117
Destination IP : 192.168.1.1

Result :

Thus the Program code using Raw sockets to implement Packet sniffing was executed successfully.